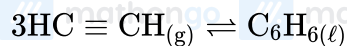


Q1: 24 Feb (Shift 2) - Numerical

Assuming ideal behaviour, the magnitude of $\log K$ for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is _____. (Integer answer)

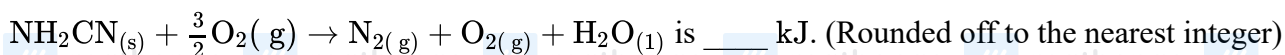


[Given : $\Delta_f G^\circ(\text{HC} \equiv \text{CH}) = -2.04 \times 10^5 \text{ J mol}^{-1}$; $\Delta_f G^\circ(\text{C}_6\text{H}_6) = -1.24 \times 10^5 \text{ J mol}^{-1}$

$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

Q2: 25 Feb (Shift 1) - Numerical

The reaction of cyanamide, $\text{NH}_2\text{CN}_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be $-742.24 \text{ kJ mol}^{-1}$. The magnitude of ΔH_{298} for the reaction



[Assume ideal gases and $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

Q3: 25 Feb (Shift 2) - Numerical

Five moles of an ideal gas at 293 K is expanded isothermally from an initial pressure of 2.1 MPa to 1.3 MPa against a constant external 4.3 MPa . The heat transferred in this process is _____ kJ mol^{-1} . (Rounded-off to the nearest integer) [Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

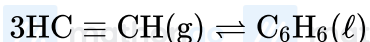
Q4: 26 Feb (Shift 1) - Numerical

For a chemical reaction $A + B \rightleftharpoons C + D$ ($\Delta_r H^\ominus = 80 \text{ kJ mol}^{-1}$) the entropy change $\Delta_r S^\ominus$ depends on the temperature T (in K) as $\Delta_r S^\ominus = 2T \left(\text{J K}^{-1} \text{ mol}^{-1} \right)$

Minimum temperature at which it will become spontaneous is _____ K .

Answer Key**Q1 (855)****Q2 (741)****Q3 (15)****Q4 (200)**

Q1 (855)



$$\Delta G_r^\circ = \Delta G_f^\circ [\text{C}_6\text{H}_6(\ell)] - 3 \times \Delta G_f^\circ [\text{HC} \equiv \text{CH}]$$

$$= [-1.24 \times 10^5 - 3 \times (-2.04 \times 10^5)]$$

$$= 4.88 \times 10^5 \text{ J/mol}$$

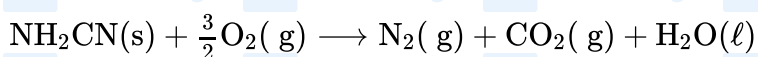
$$\Delta G_r^\circ = -RT \ln(K_{\text{eq}})$$

$$\log(K_{\text{eq}}) = \frac{-\Delta G_r^\circ}{2.303RT}$$

$$= \frac{-4.88 \times 10^5}{2.303 \times 8.314 \times 298}$$

$$= -8.55 \times 10^1 = 855 \times 10^{-1}$$

Q2 (741)



$$\Delta n_{\text{g}} = (1 + 1) - \frac{3}{2} = \frac{1}{2}$$

$$\Delta H = \Delta U + \Delta n_{\text{g}}RT$$

$$= -742.24 + \frac{1}{2} \times \frac{8.314 \times 298}{1000}$$

$$= -742.24 + 1.24$$

$$= 741 \text{ kJ/mol}$$

Q3 (15)

$$\text{Moles } (n) = 5$$

$$T = 293\text{K}$$

Process = IsoT. $\rightarrow$ Irreversible

$$P_{\text{ini}} = 2.1 \text{ MPa}$$

$$P_t = 1.3 \text{ MPa}$$

$$P_{\text{ext}} = 4.3 \text{ mPa}$$

$$\text{Work} = -P_{\text{ext}} \Delta v$$

$$= -4.3 \times \left(\frac{5 \times 293R}{1.3} - \frac{5 \times 293}{2.1} \right)$$

$$= -5 \times 293 \times 8.314 \times 43 \left(\frac{1}{13} - \frac{1}{21} \right)$$

Hints and Solutions

MathonGo

$$= \frac{5 \times 293 \times 8.314 \times 43 \times 8}{21 \times 13}$$

$$= -15347.7049 \text{ J} = -15.34 \text{ KJ}$$

Isothermal process, so $\Delta U = 0$ w = -Q

$$Q = 15.34 \text{ KJ/mol}$$

So answer is 15

Q4 (200)

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

To make the process spontaneous

$$\Delta G^\circ < 0$$

$$\Delta H^\circ - T\Delta S^\circ < 0$$

$$T > \frac{\Delta H^\circ}{\Delta S^\circ}$$

$$T > \frac{80000}{2 T}$$

$$2T^2 > 80000$$

$$T^2 > 40000$$

$$T > 200$$

The minimum temperature to make it spontaneous is 200 K.